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Research focus: galaxy formation and evolution across cosmic time with multiwavelength observations

EDUCATION

Texas A&M University <i>Ph.D. Astronomy</i> Advisor: Dr. Justin Spilker Thesis: Tracing Galaxy Evolution Across Cosmic Time with Post-Starburst Galaxies	College Station, TX <i>Expected May 2027</i>
<i>M.S. Astronomy</i> GPA: 3.92/4.00 Project: Quenching via Tidal Removal of Cold Gas in Intermediate-z Massive Post-Starburst Galaxies	<i>May 2025</i>
Michigan State University <i>B.S. Astrophysics</i> GPA: 3.92/4.00 Majors: Astrophysics, Physics	East Lansing, MI <i>May 2022</i>

RESEARCH POSITIONS/EXPERIENCE

Graduate Student (Advisor: Dr. Justin Spilker) <i>Texas A&M University</i>	August 2022 – Present <i>College Station, TX</i>
Student Research Assistant (Advisor: Dr. Laura Chomiuk) <i>Michigan State University Observatory Research Program</i>	August 2021 – May 2022 <i>East Lansing, MI</i>
NSF REU Intern (Advisor: Dr. Juergen Ott) <i>National Radio Astronomy Observatory</i>	May 2021 – October 2021 <i>Socorro, NM (Remote)</i>
Student Research Assistant (Advisor: Dr. Kaitlin Cook) <i>National Superconducting Cyclotron Laboratory</i>	June 2020 – May 2022 <i>East Lansing, MI</i>
Research Assistant (Advisor: Dr. Daniel Rathbun) <i>Bionics and Vision Lab, Henry Ford Health System</i>	April 2019 – August 2019 <i>Detroit, MI</i>

HONORS AND AWARDS

TAMU Dept. of Physics & Astronomy Travel Award, \$1k	January 2026
NRAO Student Observing Support Program Fellowship, \$40k	November 2024
NSF Graduate Research Fellowship Program, Honorable Mention	April 2024
Texas Space Grant Consortium Graduate Fellowship, \$5k	December 2023
NSF Research Experiences for Undergraduates, ~\$7k	June 2021
MSU College of Natural Science Undergraduate Research Support Scholarship, \$1k	January 2021

PUBLICATIONS

First-Author

3. **D’Onofrio, V.**, et al. (2026). “Timing the Onset of Radio-Mode Feedback with $z \sim 0.7$ Post-Starbursts.” In preparation.
2. **D’Onofrio, V.**, et al. (2026). “Molecular Gas Excitation in $z \sim 0.7$ Gas-Rich Post-starburst Galaxies from SQuIGGLE.” *ApJ*, 1003, 52.
1. **D’Onofrio, V.**, et al. (2025). “Quenching Through Tidal Gas Removal: Molecular Gas and Star Formation in Tidal Tails of $z \sim 0.7$ Post-Starburst Galaxies.” *ApJ*, 990, 166.

Co-Author

8. Knowlton, M., et al. (**incl. D’Onofrio, V.**) (2026). “A Possible Protocluster of Galaxies Serendipitously Discovered in the Field of an Intermediate-redshift Post-starburst Galaxy.” *RNAAS*, 10, 102.
7. Luo, Y., et al. (**incl. D’Onofrio, V.**) (2026). “A Multiwavelength Evaluation of AGN in the Post-Starburst Phase.” *ApJ*, 1000, 24.
6. Setton, D., et al. (**incl. D’Onofrio, V.**) (2025). “SQuIGGLE: Buried star formation cannot explain the rapidly fading CO(2–1) luminosity in massive, $z \sim 0.7$ post-starburst galaxies.” *AJ*, 170, 351.
5. Spilker, J., et al. (**incl. D’Onofrio, V.**) (2025). “Unusually High Gas-to-Dust Ratios Observed in High-Redshift Quiescent Galaxies.” *ApJL*, 993, L40.
4. Kumar, A., et al. (**incl. D’Onofrio, V.**) (2025). “Meet the Neighbors: Gas Rich “Buddy Galaxies” are Common Around Recently Quenched Massive Galaxies in the SQuIGGLE Survey.” *RNAAS*, 9, 243.
3. Suess, K., et al. (**incl. D’Onofrio, V.**) (2025). “Cold gas in a post-starburst pair at $z \sim 1.4$: major mergers as a pathway to quenching in the HeavyMetal survey.” *ApJ*, 993, 158.
2. Zhu, P., et al. (**incl. D’Onofrio, V.**) (2025). “SQuIGGLE: Observational Evidence of Low Ongoing Star Formation Rates in Gas-rich Post-starburst Galaxies.” *ApJ*, 981, 60.
1. Verrico, M., et al. (**incl. D’Onofrio, V.**) (2023). “Merger Signatures are Common, but not Universal, in Massive, Recently Quenched Galaxies at $z \sim 0.7$.” *ApJ*, 949, 5.

AWARDED PROPOSALS

Principal Investigator

5. **ALMA Cycle 13** - 2026.1.00767.S (21.9 hours): The Fate of Molecular Gas After the Burst: Tracing the Dense Gas Content in $z \sim 0.7$ Post-Starburst Galaxies (PI: **V. D’Onofrio**)
4. **VLA Semester 26A** - VLA/26A-427 (21.0 hours): What Powers the Radio? Morphological Constraints in High- z Post-Starbursts (PI: **V. D’Onofrio**)
3. **ALMA Cycle 12** - 2025.1.01006.S (12.7 hours): What’s Left Behind: A Census of the Cold ISM in the First Massive Quiescent Galaxies (PI: **V. D’Onofrio**)
2. **ALMA Cycle 11** - 2024.1.01252.S (9.8 hours): Diffuse or Dense: Probing the Physical State of Massive Gas Reservoirs in $z \sim 0.7$ Quenched Galaxies (PI: **V. D’Onofrio**)
1. **VLA Semester 24B** - VLA/24B-451 (21.0 hours): Measuring Jet Ages to Test Radio AGN Feedback with Massive $z \sim 0.7$ Post-Starbursts (PI: **V. D’Onofrio**)

Co-Investigator

10. **ALMA Cycle 13** - 2026.1.01631.S (13.2 hours): Cracking the depletion time problem in CO-luminous post-starburst galaxies: 13CO as a test of the CO-to-H₂ conversion (PI: D. Setton)
9. **ALMA Cycle 13** - 2026.1.01360.S (16.7 hours): Quenching, Fast and Slow: Molecular Gas in Green Valley Galaxies at $z \sim 0.6$ (PI: K. Suess)
8. **ALMA Cycle 13** - 2026.1.00570.S (30.3 hours): [CI]+CO Diagnostics of the Circumgalactic Medium of GS-9209, a Massive Quiescent Galaxy at $z=4.658$ (PI: F. Sun)
7. **Roman Cycle 1** - Proposal ID: 10244, Program ID: 19068 (Program Category: Small): Connecting Morphological Transformation to Rapid Quenching: Resolving the Structure of $z \sim 1$ Post-Starburst Galaxies with Roman (PI: J. Spilker)
6. **VLA Semester 26A** - VLA/26A-430 (45.0 hours): Are High-Redshift Quiescent Galaxies Gas-Rich but Dust-Poor? (PI: J. Spilker)
5. **ALMA Cycle 12** - 2025.1.01067.S (45.6 hours): Dust Rising from the ASHES: A Comprehensive ALMA Survey of Remnant Dust in the Earliest Quiescent Galaxies (PI: Z. Ji)
4. **ALMA Cycle 11 w/ JWST Cycle 3** - 2024.1.01064.S/JWST-GO-06719 (15.5/4.7 hours): Mapping Cold Gas and Star Formation in Gas Rich Post-Starburst Galaxies Near Cosmic Noon (PI: D. Setton)
3. **ALMA Cycle 11** - 2024.1.00216.S (13.1 hours): Timing the Onset of Unexpected Dust Destruction using High-Redshift Post-Starburst Galaxies (PI: J. Spilker)
2. **ALMA Cycle 10** - 2023.1.01012.S (12.1 hours): Does Molecular Gas Survive Quenching Near Cosmic Noon? (PI: D. Setton)
1. **ALMA Cycle 10** - 2023.1.00948.S (13.1 hours): Timing the Onset of Unexpected Dust Destruction using High-Redshift Post-Starburst Galaxies (PI: J. Spilker)

TALKS

Invited

1. **D’Onofrio, V.**, et al. “Tidal Removal of Cold Molecular Gas in $z \sim 0.7$ Post-Starburst Galaxies: A Qualitatively New Quenching Mechanism,” Texas A&M Nuclear-Astro Seminar, College Station, TX, May 24, 2024.

Contributed

10. **D’Onofrio, V.**, et al. “Insights from the Radio into Star Formation Suppression in SQuIGGLE Post-Starburst Galaxies,” Texas A&M Astrosymposium, College Station, TX, August 21, 2026.
9. **D’Onofrio, V.**, et al. “Investigating Star Formation Suppression via Higher-Redshift Post-Starburst Galaxies,” The End of Star Formation, Urbana-Champaign, IL, March 3, 2026.
8. **D’Onofrio, V.**, et al. “Investigating Star Formation Suppression via the ISM of Higher-Redshift Post-Starburst Galaxies,” Decoding Galactic Evolution through the Interplay of the Multi-Phase Interstellar Medium, Nagoya, August 25, 2025.

7. **D’Onofrio, V.**, et al. “Tracing galaxy quenching via post-starbursts with recent observations from the radio,” Texas A&M Astrosymposium, College Station, TX, August 18, 2025.
6. **D’Onofrio, V.**, et al. “Probing Star Formation Suppression with Higher-z Post-Starbursts,” Texas A&M Astrosymposium, College Station, TX, August 15, 2024.
5. **D’Onofrio, V.**, et al. “Tidal Removal of Cold Molecular Gas in $z \sim 0.7$ Post-Starburst Galaxies: A Qualitatively New Quenching Mechanism,” McMaster University Star Formation Workshop, Hamilton, ON, August 14, 2024.
4. **D’Onofrio, V.**, et al. “Tidal Removal of Cold Molecular Gas in $z \sim 0.7$ Post-Starburst Galaxies: A Qualitatively New Quenching Mechanism,” STScI Spring Symposium 2024, Baltimore, MD, April 19, 2024.
3. **D’Onofrio, V.**, et al. “Quenching in Intermediate-Redshift Massive Post-Starburst Galaxies,” Texas A&M Astrosymposium, College Station, TX, August 17, 2023.
2. **D’Onofrio, V.**, et al. “Molecular Gas Toward Sagittarius B2: The Galactic Center’s Complex Giant Molecular Cloud,” Texas A&M Astrosymposium, College Station, TX, August 18, 2022.
1. **D’Onofrio, V.**, et al. “Characterizing Molecular Gas Towards Sagittarius B2: The Galactic Center’s Complex Giant Molecular Cloud,” 2021 NRAO/GBO Summer Student Symposium, virtual participation, August 13, 2021.

Public

2. **D’Onofrio, V.** “A Day in the Life of an Astronomy PhD Student,” TAMU High School Outreach Event, College Station, TX, November 12, 2025.
1. **D’Onofrio, V.** “Why You Shouldn’t Go to Mars,” Astronomy on Tap, Bryan, TX, August 23, 2023.

Other

4. **D’Onofrio, V.**, et al. “Evolution Through the Post-starburst Phase,” Texas A&M Exgal Group Meeting, College Station, TX, October 20, 2025.
3. **D’Onofrio, V.**, et al. “Tidal Removal of Cold Molecular Gas in $z \sim 0.7$ Post-Starburst Galaxies: A Qualitatively New Quenching Mechanism,” Master’s/Preliminary Exam Defense, College Station, TX, October 28, 2024.
2. **D’Onofrio, V.**, et al. “Star Formation Quenching in Intermediate-Redshift Massive Post-Starburst Galaxies,” Texas A&M Exgal Group Meeting, College Station, TX, April 11, 2024.
1. **D’Onofrio, V.**, et al. “Molecular Gas Toward Sagittarius B2: The Galactic Center’s Complex Giant Molecular Cloud,” MSU Society of Physics Students Coffee Breaks, East Lansing, MI, November 11, 2021.

LEADERSHIP & OUTREACH

STEM Pen Pal

Letters to a Pre-Scientist

August 2025 – Present

Texas A&M University

Coordinator

Mentoring and Advising Graduates in an Inclusive Community (MAGIC)

February 2025 – Present

Texas A&M University

Coordinator <i>Graduates Learning Astro and Soft Skills (GLASS)</i>	October 2023 – February 2025 <i>Texas A&M University</i>
<i>In the News</i> Presenter <i>Astronomy on Tap</i>	October 2023 – April 2026 <i>Bryan, TX</i>
Mentor <i>Mentoring and Advising Graduates in an Inclusive Community (MAGIC)</i>	August 2023 – August 2026 <i>Texas A&M University</i>
Trivia Creator <i>Astronomy on Tap</i>	May 2023 – July 2024 <i>Bryan, TX</i>
Astronomer/Physicist <i>Adopt-a-Physicist</i>	October 2023 – Present <i>Texas A&M University</i>

TEACHING EXPERIENCE

Teaching Assistant, Lab Instructor <i>ASTR 111: Overview of Modern Astronomy</i>	January 2024 – May 2024 <i>Texas A&M University</i>
Teaching Assistant <i>ASTR 320: Astrophysical Research Methods</i>	August 2023 – December 2023 <i>Texas A&M University</i>
Teaching Assistant, Lab Instructor <i>ASTR 111: Overview of Modern Astronomy</i>	January 2023 – May 2023 <i>Texas A&M University</i>
Teaching Assistant <i>ASTR 401: Stars and Extrasolar Planets</i>	August 2022 – December 2022 <i>Texas A&M University</i>

OBSERVATION EXPERIENCE

MSU Observatory 0.6-meter Telescope, **30 hrs**